MAT 534 HW06

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Problem 1. Let R be a ring with 1. Show that if $0 = 1$, then R is the zero ring.

Solution. Let $x \in R$. Then, using the result of Problem 2a,

$$x = 1 \cdot x = 0 \cdot x = 0$$

so R has exactly one element and is the zero ring. $\square$

Problem 2. Let R be a ring. We use $-a$ to denote the additive inverse of $a \in R$.

(a) Prove that $0 \cdot a = 0$ for every $a \in R$.

(b) Prove that $(-a)(-b) = ab$ for every $a, b \in R$.

Solution. (a) Since $0 + 0 = 0$,

$$0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a,$$

and since R is a group under $+$, we can cancel $0 \cdot a$ from both sides to get $0 = 0 \cdot a$.

(b) First, we compute

$$0 = 0 \cdot y = (x + (-x)) \cdot y = xy + (-x)y \implies -(xy) = (-x)y \tag{1}$$
$$0 = x \cdot 0 = x \cdot (y + (-y)) = xy + x(-y) \implies -(xy) = x(-y) \tag{2}$$

(where $0 = x \cdot (0 + 0) - x \cdot 0 = x \cdot 0$) for any $x, y \in R$. Then

$$ab = -(-(ab)) \overset{(1)}{=} -((-a)b) \overset{(2)}{=} (-a)(-b).$$ $\square$

Problem 3. Prove that every ideal in the ring $\mathbb{Z}$ is of the form (n) for some $n \in \mathbb{Z}$.

Solution. Given ideal $I \subseteq \mathbb{Z}$, let n be the smallest positive integer in I. Then, for any $m \in I$, we can apply the division algorithm to get that $m = nq + r$ for some integers q and r such that $0 \leq r < n$. Since I is a group under addition, $m - nq = r$ is in I, so $0 \leq r < n$ implies that r must be zero, since we would otherwise contradict the minimality of n; this means $m = nq$ is a multiple of n. That is, every element of I is a multiple of n, and conversely $n \in I$ means $nk \in I$ for all $k \in \mathbb{Z}$, so $I = (n)$. $\square$

Problem 4. Let F be a field, and let R be the set of all functions $a\colon \mathbb{Z}_{\geq 0} \to F$. Here we think of $a \in R$ as representing the formal power series

$$\sum_{n=0}^{\infty} a(n)x^n = a(0) + a(1)x + a(2)x^2 + \cdots$$

The ring of formal power series is usually denoted by $F[\![x]\!]$.

(a) Show that R is a commutative ring with 1, under the operations

$$(a+b)(n) = a(n) + b(n) \quad \text{and} \quad (a \cdot b)(n) = \sum_{i=0}^{n} a(i)b(n-i).$$

(b) Determine the set of units $R^\times$.

(c) Determine the ideals in the ring R.

Solution. (a) The addition operation inherits associativity from the addition of F; the zero element is the function that is constantly zero; and the inverse of $a\colon \mathbb{Z}_{\geq 0} \to F$ is $-a\colon \mathbb{Z}_{\geq 0} \to F$ defined by $(-a)(n) = -(a(n))$.

The multiplication operation can be written more symmetrically as

$$(a \cdot b)(n) = \sum_{i+j=n} a(i)b(j),$$

which lets us recover commutativity immediately because $a(i)b(j) = b(j)a(i)$ and associativity because

$$\begin{aligned}((a \cdot b) \cdot c)(n) = \sum_{m+k=n} \sum_{i+j=m} a(i)b(j)c(k) &= \sum_{i+j+k=n} a(i)b(j)c(k) \\ &= \sum_{i+m'=n} \sum_{j+k=m'} a(i)b(j)c(k) = (a \cdot (b \cdot c))(n).\end{aligned}$$

The multiplicative identity in this ring is the function $e\colon \mathbb{Z}_{\geq 0} \to F$ given by $e(0) = 1$ and $e(n) = 0$ for $n \geq 1$, since

$$(a \cdot e)(n) = \sum_{i=0}^{n} a(n-i)e(i) = a(n)e(0) + \sum_{i=1}^{n} a(n-i) \underbrace{e(i)}_{0} = a(n).$$

(b) The set of units of R is the set of elements a with $a(0) \neq 0$. To check that all such elements are invertible, we can construct an inverse b as such:

- Set $b(0) = a(0)^{-1}$, so $(a \cdot b)(0) = 1$.
- For $n \geq 1$, we want

  $$(a \cdot b)(n) = \sum_{i=0}^{n} a(n-i)b(i) = 0,$$

  so, solving for $b(n)$ here, we can choose

  $$b(n) := a(0)^{-1} \sum_{i=0}^{n-1} a(n-i)b(i)$$

  to force this, given $b(n-1), \ldots, b(0)$.

On the other hand, if $a(0) = 0$, then for any $b \in R$ $(a \cdot b)(0) = a(0)b(0) = 0 \neq 1$, so a cannot be invertible. □

Problem 5. Find a ring R_0 in ⟮unital commutative rings⟯ such that, for any unital commutative ring R, the set $\mathrm{Hom}(R_0, R)$ is in one-to-one correspondence with R.

Solution. Let $R_0 = \mathbb{Z}[x]$. Then a morphism $\phi\colon R_0 \to R$ is uniquely determined by $\phi(x)$ (since $\phi(1) = 1$ is fixed), and $\phi(x)$ can be set to any element of R, so there is a morphism for every element of R. □

Problem 6. Find a ring R_1 in $(\!(\text{unital commutative rings})\!)$ such that, for any unital commutative ring R, the set $\mathrm{Hom}(R_1, R)$ is in one-to-one correspondence with $R^\times$.

Solution. Let $R_1 = \mathbb{Z}[x, 1/x]$ be the ring of Laurent polynomials over $\mathbb{Z}$. Then a morphism $\phi\colon R_1 \to R$ is uniquely determined by $\phi(x)$ (since $\phi(1) = 1$ is fixed). Let ϕ_r be the morphism sending x to $r \in R$. On one hand r must be a unit for ϕ_r to be well-defined because $\phi_r(x)\phi_r(x^{-1}) = \phi_r(1) = 1$ so $\phi_r(x) = r$ has an inverse; on the other hand, if r is a unit then ϕ_r is well-defined, being the evaluation morphism $p(x) \mapsto p(r)$; thus $r \mapsto \phi_r$ is a bijection from R to $\mathrm{Hom}(R_1, R)$. □

Problem 7. If R is a unital ring, the set $C(R) = \{x \in R : xy = yx \text{ for all } y \in R\}$ is called the **center** of R. Elements of $C(R)$ are called **central**.

(a) Show that $C(R)$ is a subring of R.

(b) Let $R = M_n(F)$ be the ring of $n \times n$ matrices over field F. What is the center of R?

(c) Let G be a finite group, and let R be the integral group ring $\mathbb{Z}[G]$ of G. What is the center of R?

Solution. (a) Clearly 1 is central. If a and b are central then for all $x \in R$, $(a + b)x = ax + bx = xa + xb = x(a + b)$ so $a + b$ is central, and $abx = axb = xab$ implies ab is central. These three statements imply $C(R)$ is a subring of R.

(b) The center of R is the diagonal matrices λI, $\lambda \in F$ (where I is the identity matrix). Clearly these commute with all elements of R, so we need to show that any central element is of this form.

Let $E_{ij} \in R$ be the matrix whose entries are all zero except for a 1 in row i, column j. For any $A \in R$, $E_{ij}A$ is the matrix whose ith row is equal to the jth row of A and all other entries are zero, while AE_{ij} is the matrix whose jth column is equal to the ith column of A and all other entries are zero.

If A is central in R, then these two matrices are equal for all $i, j \in \{1, \ldots, n\}$. The entry at row i, column j in the product is equal to both the entry at row j, column j and the entry at row i, column i in A, meaning as i and j vary we see that A has the same element of F at every entry in the diagonal. Moreover, if $i \neq j$ then the entry of $E_{ji}A$ at row j, column j is equal to the row i, column j entry of A, and the corresponding entry of AE_{ji} is equal to zero since the nonzero row is row i; therefore the entry in A at row i, column j is zero, so any entry that is off the main diagonal must be zero, i.e. A is a multiple of I.

(c) Let C_1, C_2, $\ldots$, C_n be the conjugacy classes of G. Then let $s_j = \sum_{g \in C_j} g$ and let

$$Z = \left\{ \sum_{j=1}^{n} c_j s_j : c_1, \ldots, c_n \in \mathbb{Z} \right\}.$$

It is easy to see that $Z \subseteq C(\mathbb{Z}[G])$, since if $h \in G$ then $hC_j = C_j h$, so that $hs_j = s_j h$. Conversely, if $\sum_{x \in G} c_x x$ is central, then, conjugating by g,

$$\sum_{x \in G} c_x x = \sum_{x \in G} c_x gxg^{-1} = \sum_{x \in G} c_{g^{-1}xg} x,$$

so if y is conjugate to x then $c_y = c_x$, i.e. the coefficients of any central element are constant on conjugacy classes, so the central element is in Z. □

Problem 8. Suppose that $f(x, y)$ is a polynomial in $\mathbb{Z}[x, y]$, and $R = \mathbb{Z}[x, y]/(f(x, y))$. Show that $\mathrm{Hom}(R, \mathbb{R})$ is in one-to-one correspondence with the points on the graph of $f(x, y) = 0$ in $\mathbb{R}^2$.

Solution. $\operatorname{Hom}(R,\mathbb{R})$ is in one-to-one correspondence with the maps $\phi\colon \mathbb{Z}[x,y] \to \mathbb{R}$ such that $(f(x,y)) \subseteq \ker\phi$. Let $\phi_{r,s}$ be the ring homomorphism $\mathbb{Z}[x,y] \to \mathbb{R}$ sending x to r and y to s; the kernel of this map contains (f) if and only if $\phi_{r,s}(f(x,y)) = f(\phi_{r,s}(x), \phi_{r,s}(y)) = f(r,s) = 0$, which gives the bijection between points $(r,s) \in f^{-1}(0)$ and morphisms $\phi_{r,s} \in \operatorname{Hom}(R,\mathbb{R})$. $\square$

Problem 9. Let R be a unital ring.

(a) If $e \in R$ is a central idempotent, show that $R \cong R/(e) \times R/(1-e)$.

(b) Let H be a subgroup of a finite group G. Show that the element $e_H = \frac{1}{|H|}\sum_{g\in H} g$ is an idempotent in the group ring $\mathbb{Q}[G]$. Under what conditions on H does e_H belong to the center of the group ring?

Solution. (a) If e is a central idempotent then so is $1-e$. Clearly $(e)+(1-e) = (1)$ and the intersection $(e)\cap(1-e)$ is trivial since if $ae = b(1-e)$ for some $a,b \in R$ then this common element is $ae = ae^2 = b(1-e)e = 0$. The Chinese remainder theorem then implies that $R \cong R/(e)\times R/(1-e)$.

(b) We can compute

$$e_H^2 = \frac{1}{|H|^2}\sum_{x\in H}\sum_{g\in H} xg = \frac{1}{|H|^2}\cdot|H|\cdot\sum_{g\in H} g = \frac{1}{|H|}\sum_{g\in H} g = e_H$$

since $g \mapsto xg$ is a bijection on H. By the same logic to 7(c) (the group ring being over $\mathbb{Q}$ and not $\mathbb{Z}$ changes nothing), e_H is central if and only if H is a union of conjugacy classes of G, i.e. H is normal. $\square$

Problem 10. Let X be a compact Hausdorff space, and let $C(X)$ be the ring of continuous functions $f\colon X \to \mathbb{R}$.

(a) Is $C(X)$ an integral domain?

(b) Let $x \in X$ be any point. Show that $\{f \in C(X) : f(x) = 0\}$ is a maximal ideal of $C(X)$.

(c) Show that every maximal ideal of $C(X)$ is of this form.

Solution. (a) No; let $A \subsetneq X$ be closed with nonempty interior and $B = \overline{X\setminus A} \neq X$ also be closed, so $A\cup B = X$. Then, $C(X)$ contains a nonzero function that is zero on A (concrete construction: pick a point $p \notin A$ and, since compact Hausdorff spaces are normal, use Urysohn's lemma to get a function that's zero on A and 1 at p) and a nonzero function that is zero on B. The product of these two functions is zero on $A\cup B$, so the two functions are zero divisors in $C(X)$.

(b) Let $M_x = \{f \in C(X) : f(x) = 0\}$, and let $g \in C(X)$ be a function that is not zero at x. Then, $B = g^{-1}(0) \neq X$ is a closed subset of X. Since X is compact Hausdorff, it is normal, so there exists, by Urysohn's lemma, a continuous function $f\colon X \to \mathbb{R}$ such that $f|_A = 0$ and $f(p) = 1$ where p is a point in $X\setminus A$. Then, f^2+g^2 is a continuous function that is nowhere zero, which means it is a unit in $C(X)$, as it has inverse $\frac{1}{f^2+g^2}$. That is, the ideal generated by M_x and any g outside M_x is equal to the entire ring $C(X)$, so M_x must be maximal.

(c) Let M be an ideal. If, for each $x \in X$, M contains a function that is nonzero at x, then the sets $f^{-1}(\mathbb{R}\setminus\{0\})$ are an open cover of X, so we can choose finitely many of these functions $f_1,\dots,f_n$ where $f_1^{-1}(\mathbb{R}\setminus\{0\})\cup\cdots\cup f_n^{-1}(\mathbb{R}\setminus\{0\}) = X$. This means $f_1^2+\cdots+f_n^2 \in M$ is nowhere zero, so M contains a unit, hence $M = C(X)$.

Thus, if M is maximal, there exists a closed subset $A \subseteq X$ such that $f|_A = 0$ for all $f \in M$. However this ideal is contained in M_x for any $x \in A$, so the M_x are the only such ideals that are maximal. $\square$

Problem 11. Let R be a unital commutative ring. Show that an element $r \in R$ is a unit if and only if r is not contained in any maximal ideal of R.

Solution. If r is a unit then an ideal containing r is the entire ring, so it cannot be maximal.

If r is not a unit then consider the set of all proper ideals of R that contain r, ordered by inclusion. Given a totally ordered subcollection of these ideals $\{I_\alpha\}_\alpha$, the union $\bigcup_\alpha I_\alpha$ is also an ideal, and this ideal is not R because none of the I_α contains a unit, so it is an upper bound for the totally ordered subcollection. By Zorn's lemma, the collection of proper ideals containing r has a maximal element, which is a maximal ideal containing r. $\square$